1. Introduction

In general relativity, gravity is described as the curvature of spacetime. Massive objects bend spacetime, and photons follow null geodesics in this curved geometry. As a result, light paths near compact objects are strongly deflected, producing gravitational lensing.

Black holes are ideal laboratories for studying such relativistic effects. Phenomena such as the black hole shadow, photon ring-like structures, strong lensing, and relativistic disk emission arise from photon motion in the strong-gravity region. Motivated by these features, we perform ray tracing simulations of Schwarzschild and Kerr black holes with accretion disks.

Using intensity maps, redshift maps, and selected three-dimensional photon trajectories, we qualitatively interpret the black hole shadow, photon ring-like features, gravitational lensing, ISCO-related disk structures, and redshift/Doppler brightness asymmetry.

2. Method: Null Geodesic Ray Tracing

In curved spacetime, photons move along null geodesics. The null condition and geodesic equation are given by

$$g_{\mu\nu}P^\mu P^\nu = -m^2 = 0 \qquad \cdots(1)$$

$$\frac{dP^\mu}{d\lambda} + \Gamma^\mu{}_{\alpha\beta}P^\alpha P^\beta = 0 \qquad \cdots(2)$$

where $P^\mu = dx^\mu/d\lambda$ and λ is an affine parameter. We use the $(-,+,+,+)$ metric signature. The spacetime geometry is encoded in the metric tensor $g_{\mu\nu}$, and it appears in the geodesic equation through the Christoffel symbol $\Gamma^\mu{}_{\alpha\beta}$.

Ray tracing is a computational method for calculating the paths of light in curved spacetime. In backward ray tracing, photon trajectories are integrated backward from the observer screen by solving Eqn. 2 numerically. Each pixel on the screen corresponds to an initial photon direction, and the ray is traced to determine whether it intersects the accretion disk, falls into the event horizon, or escapes without contributing to the image. This allows us to construct observable images from the underlying spacetime and emission model. Ray-traced images can also be compared with astronomical observations, such as EHT images, to test whether observed black hole features are consistent with GR-based models.

In this project, we developed a ray tracing simulator for Schwarzschild and Kerr black holes with accretion disks. The Schwarzschild black hole is the simplest non-rotating black hole solution, characterized only by its mass M. For the Schwarzschild case, the spacetime is described by

$$ds^2 = -\left(1-\frac{2M}{r}\right)dt^2 + \left(1-\frac{2M}{r}\right)^{-1}dr^2 + r^2 d\Omega^2.$$

The Kerr black hole describes a rotating black hole with mass M and spin parameter $a \equiv J/M$, introducing frame dragging and spin-dependent changes in photon trajectories and disk inner radius. In our project, Kerr metric in Boyer-Lindquist coordinates is used for rotating black holes.

The observer position, disk inclination, and image resolution are treated as observer parameters. The observer screen is placed at a finite distance from the black hole. Each pixel corresponds to an initial photon direction in the observer's local tetrad frame. The photon four-momentum is transformed from the local frame to the global coordinates and then integrated backward using the 4th-order Runge-Kutta method (RK4). The fate of each photon determines the value of the corresponding image pixel.

The accretion disk is modeled as a geometrically thin disk on the equatorial plane. A photon contributes to the image when its trajectory intersects the disk within the prescribed inner and outer radii. In the simple disk model, the emitted intensity is prescribed phenomenologically as

$$I_{\rm em} \propto r^{-q}$$

In contrast, the Novikov-Thorne disk is a relativistic thin disk model in which the disk flux profile is determined by the black hole spacetime, and the inner edge is typically located at the ISCO. Thus, the simple disk is useful for qualitative visualization, while the Novikov-Thorne disk provides a more physically motivated model for the inner accretion disk.

For each disk-intersecting photon, the redshift factor is computed as

$$g \equiv \frac{\nu_{\rm obs}}{\nu_{\rm em}} = \frac{(P_\mu U^\mu)_{\rm obs}}{(P_\mu U^\mu)_{\rm em}} \qquad \cdots(3)$$

$$I_{\nu,\rm obs} = g^3 I_{\nu,\rm em} \qquad \cdots(4)$$

where U^μ is the four-velocity of the observer or emitter. Using the disk emission model and the redshift factor, we obtain the observed intensity map and redshift map.

Finally, the null condition is used as a numerical consistency check. The Levi-Civita connection is torsion-free and metric-compatible,

$$\nabla_\alpha g_{\mu\nu} = 0$$

Together with the geodesic equation, this implies that the mass-shell condition is conserved along an affinely parametrized geodesic:

$$\begin{aligned}\frac{d}{d\lambda}(g_{\mu\nu}P^\mu P^\nu) &= \nabla_P(g_{\mu\nu}P^\mu P^\nu)\\ &= (\nabla_P g_{\mu\nu})P^\mu P^\nu + 2g_{\mu\nu}(\nabla_P P)^\mu P^\nu\\ &= 0 \qquad \cdots(5)\end{aligned}$$

Therefore, a photon that is initially null remains null in the exact theory. In the numerical simulation, we monitor the deviation of $g_{\mu\nu}P^\mu P^\nu$ from zero. If this error remains small, the computed trajectories can be regarded as good approximations to true null geodesics.

3. Result 1 : Relativistic Phenomena in Ray-Traced Images

In this section, we first focus on the Schwarzschild black hole as a reference case. Since the Schwarzschild spacetime is non-rotating and spherically symmetric, it provides a simpler setting for identifying the basic relativistic features in ray-traced images. The effects of spin and Kerr geometry will be discussed later through qualitative comparisons in Section 4.

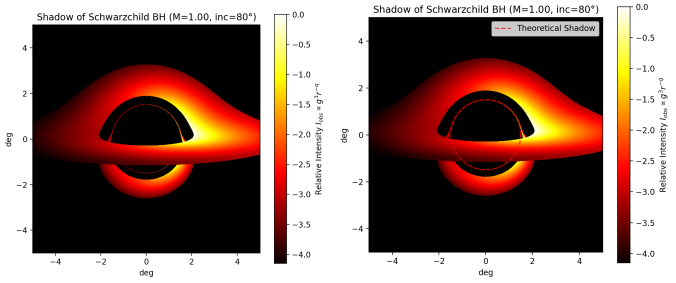


Fig. 1. Schwarzschild ray-traced image and theoretical shadow guide.

3.1. Black hole shadow

In Figure 1, the central dark region corresponds to the black hole shadow. The shadow is not a direct image of the event horizon. Rather, it is the apparent dark region on the observer's image plane formed by photon trajectories that are captured by the black hole.

The event horizon is a causal boundary in spacetime. For a Schwarzschild black hole, it is located at $r_{\rm H} = 2M$. Once a photon crosses this radius, it cannot escape to infinity. However, the apparent size of the shadow is larger than the event horizon because light rays are strongly bent by the curved spacetime. The boundary of the shadow is determined by critical null geodesics associated with the unstable photon sphere at $r_{\rm ph} = 3M$.

For a distant observer, this gives the critical impact parameter $b_{\rm crit} = 3\sqrt{3}M$. Therefore, the apparent radius of the Schwarzschild black hole shadow is $R_{\rm sh} = 3\sqrt{3}M$, which is larger than the event horizon radius ($2M$). In Figure 2, the dashed red circle shows this theoretical shadow radius. The simulated dark central region agrees well with the theoretical prediction, confirming that the ray-tracing calculation correctly reproduces the photon capture region of the Schwarzschild spacetime.

It should be noted that dark regions outside the theoretical shadow boundary are not necessarily part of the shadow; they

can also arise from the disk emission model, as discussed in Section 3.3.

3.2. Photon sphere, photon ring, and strong lensing

The photon sphere is a characteristic structure of the Schwarzschild spacetime. It is the radius at which photons can have circular null geodesics. To see this quantitatively, we consider the radial motion of photons in the Schwarzschild metric. Using the conserved energy E and angular momentum L, the radial equation can be written as

$$\left(\frac{dr}{d\lambda}\right)^2 + V_{\rm eff}(r) = E^2 \qquad \cdots (6)$$

where the effective potential is

$$V_{\rm eff}(r) = \left(1 - \frac{2M}{r}\right)\frac{L^2}{r^2} \qquad \cdots (7)$$

A circular null geodesic should satisfy

$$\frac{dr}{d\lambda} = 0, \qquad \frac{dV_{\rm eff}}{dr} = 0 \qquad \cdots (8)$$

Solving this condition gives

$$r_{\rm ph} = 3M \qquad \cdots (9)$$

This radius is called the photon sphere. However, this orbit is unstable. Since $V_{\rm eff}$ has a maximum at $r = 3M$, a small perturbation causes the photon either to fall into the black hole or to escape outward. Therefore, the photon sphere acts as a critical structure separating captured and escaping photon trajectories. The corresponding critical impact parameter is

$$b_{\rm crit} = \frac{L}{E} = 3\sqrt{3}M \qquad \cdots (10)$$

For a distant observer, this gives the apparent radius of the shadow boundary on the image plane. This value should not be interpreted as a physical radius in spacetime. Rather, it is the impact parameter of critical null geodesics seen by the observer.

The strong lensing near the photon ring can be understood qualitatively from the shape of $V_{\rm eff}$. When b is close to $b_{\rm crit}$, the photon trajectory approaches the unstable maximum of the effective potential near $r = 3M$. In this region, $E^2 - V_{\rm eff}(r)$ becomes small, so the radial motion becomes slow while the angular motion continues. As a result, the photon can accumulate a large deflection angle before reaching the observer.

This is important for the formation of higher-order disk images. A direct image corresponds to photons that reach the observer after weak bending, while higher-order images correspond to photons that wind around the black hole before escaping. Since larger winding numbers require larger deflection angles, these trajectories must have impact parameters increasingly close to $b_{\rm crit}$. Therefore, the higher-order images of the accretion disk are compressed into narrow annular regions near the critical curve.

The photon ring is therefore an image-plane feature produced by higher-order lensed images of the disk, not by emission from the photon sphere itself. Instead, light emitted from the accretion disk is strongly bent near the photon sphere and mapped to a narrow region near $b = b_{\rm crit}$ on the observer's screen. In Figure 1, the thin bright structure close to the shadow boundary can be interpreted as a photon-ring-like feature, while the broader bright region outside it mainly corresponds to the direct and weakly lensed image of the accretion disk.

3.3. ISCO and disk inner radius

Another important scale in the Schwarzschild black hole image is the innermost stable circular orbit, or ISCO. Unlike the photon sphere, which is defined for null geodesics, the ISCO is defined for massive particles orbiting around the black hole. In the Schwarzschild spacetime, the ISCO is located at

$$r_{\rm ISCO} = 6M \qquad \cdots (11)$$

In our disk model, the inner radius of the accretion disk is set to the ISCO,

$$r_{\rm in} = r_{\rm ISCO} = 6M \qquad \cdots (12)$$

Therefore, the disk does not emit light from the region $r < 6M$. This is important for interpreting the dark regions in the image. The central dark region inside the theoretical shadow boundary is caused by photon capture, while dark regions outside the shadow can appear simply because the corresponding geodesics do not intersect the emitting part of the disk.

Thus, the ISCO determines where the emitting matter starts, while the photon sphere determines the critical lensing structure near the shadow boundary. This difference explains why the direct disk image does not begin exactly at the shadow boundary.

3.4. Redshift and Doppler brightness asymmetry

The brightness asymmetry of the disk is mainly controlled by the redshift factor g. As defined in Eqn. 3, $g = \nu_{\rm obs}/\nu_{\rm emit}$ measures the frequency shift between the emitter and the observer. Since the observed intensity depends on g as shown in Eqn. 4, regions with larger g appear brighter, while regions with smaller g appear dimmer.

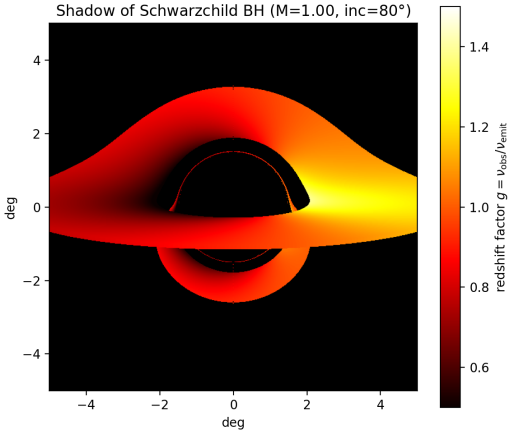


Fig. 2. Redshift factor map of the Schwarzschild disk image.

Figure 2 shows the redshift factor map of the ray-traced image. The right side of the disk has larger g, corresponding to Doppler blueshift, while the left side has smaller g, corresponding to Doppler redshift. This directly explains the left-right brightness asymmetry in Figure 1.

This asymmetry is caused by the orbital motion of the disk material. Although the Schwarzschild black hole itself has no spin, the accretion disk is rotating. Therefore, one side of the disk moves toward the observer and becomes blueshifted, while the opposite side moves away and becomes redshifted. Since the inclination is high, $i = 80°$, the line-of-sight component of the disk velocity is large, making the Doppler contrast prominent.

Gravitational redshift also contributes, especially for photons emitted from the inner disk. However, the dominant source of the left-right contrast in Figure 2 is the Doppler shift due to disk rotation. Therefore, the image can be brightness-asymmetric even in a spherically symmetric Schwarzschild spacetime.

4. Result 2 : Qualitative Comparison

In Section 3, we used the Schwarzschild black hole as a reference case to identify the main relativistic features in ray-traced images: the black hole shadow, photon-ring-like structures, ISCO-related disk features, and redshift/Doppler brightness asymmetry. In this section, we compare these features in the Kerr case and examine how spin and inclination change the visibility of the photon ring. Here, we focus on the geometric visibility of the photon-ring segment, rather than the overall brightness contrast. In particular, we interpret the visibility of the photon ring as a geometric effect: whether the corresponding near-critical photon trajectories can escape to the observer without being blocked by the optically thick accretion disk.

4.1. Schwarzschild versus Kerr black holes

Figure 3 compares the Schwarzschild image with a Kerr black hole image under the same observer inclination and disk model. In the Schwarzschild case, the spacetime is non-rotating and spherically symmetric, so the photon sphere is symmetric and the shadow boundary is nearly circular. In contrast, the Kerr spacetime is rotating, and this rotation changes the structure of null geodesics through frame dragging.

Frame dragging means that the local inertial frames are dragged in the direction of the black hole spin. This can be described using the concept of a zero-angular-momentum observer

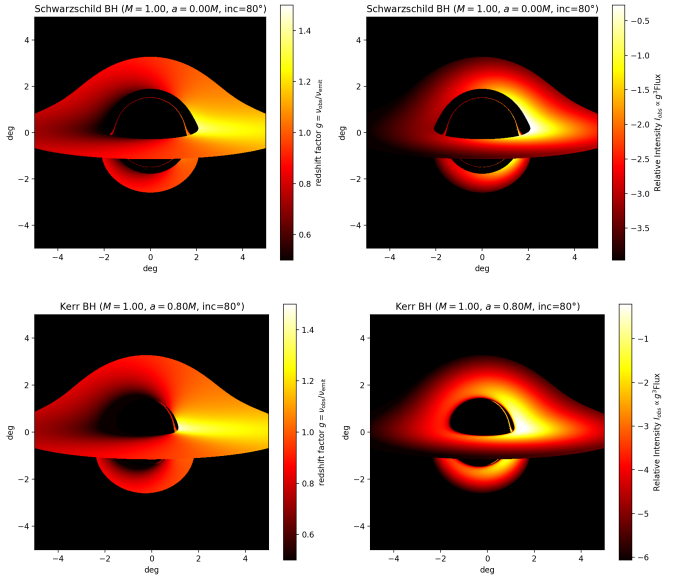


Fig. 3. Comparison between Schwarzschild and Kerr black hole images.

(ZAMO). A ZAMO has zero angular momentum with respect to infinity, but it is still forced to rotate locally because the spacetime itself is dragged by the black hole spin. Therefore, in Kerr spacetime, even the notion of a locally non-rotating frame differs from that in Schwarzschild spacetime.

In Boyer-Lindquist coordinates, the angular velocity of a ZAMO is

$$\Omega_{\rm ZAMO} = -\frac{g_{t\phi}}{g_{\phi\phi}} \tag{13}$$

For the Kerr metric, this quantity is nonzero when $a \neq 0$. This shows that even an observer with zero angular momentum is dragged in the direction of the black hole spin. Thus, frame dragging is not simply a coordinate effect; it reflects the rotation of the local inertial frame itself.

This frame dragging makes prograde and retrograde photon trajectories behave differently. Prograde photons, which move in the same direction as the black hole spin, can follow near-critical trajectories closer to the event horizon. As a result, they can be strongly bent in the deep gravitational region and escape upward through the inner region without intersecting the optically thick disk. In the following discussion, we identify the sharp crescent-shaped feature attached to the inner-right shadow boundary as the visible photon-ring segment. This feature is distinguished from the broader direct disk image by its narrow width and its location close to the near-critical shadow boundary.

Retrograde photons, on the other hand, are affected differently by frame dragging. Their near-critical trajectories occur at larger radii, and in this viewing geometry they are more likely to be intercepted by the equatorial disk. The corresponding rays tend to intersect the disk plane at larger radii before reaching the observer. Since the disk is treated as optically thick, such rays are absorbed or occluded once they hit the disk. This geometric screening explains why the photon ring is visible mainly on the prograde side, while the retrograde-side contribution is hidden by the disk.

Thus, the difference between the Schwarzschild and Kerr images is not only a change in shadow shape. Kerr spin changes the geodesic structure itself, producing an asymmetric visibility of the photon-ring segment through frame dragging and disk occlusion.

4.2. Inclination dependence

Figure 4 shows the dependence of the Kerr image on observer inclination. At high inclination, such as 60° or 80°, the observer sees the accretion disk nearly edge-on. In this geometry, the prograde and retrograde photon-ring contributions are screened differently by the disk.

On the prograde side, frame dragging allows near-critical photon trajectories to remain closer to the black hole. These photons can be bent sharply in the strong-field region and then escape through the inner region above the disk plane. Therefore, the corresponding photon-ring segment can avoid being blocked

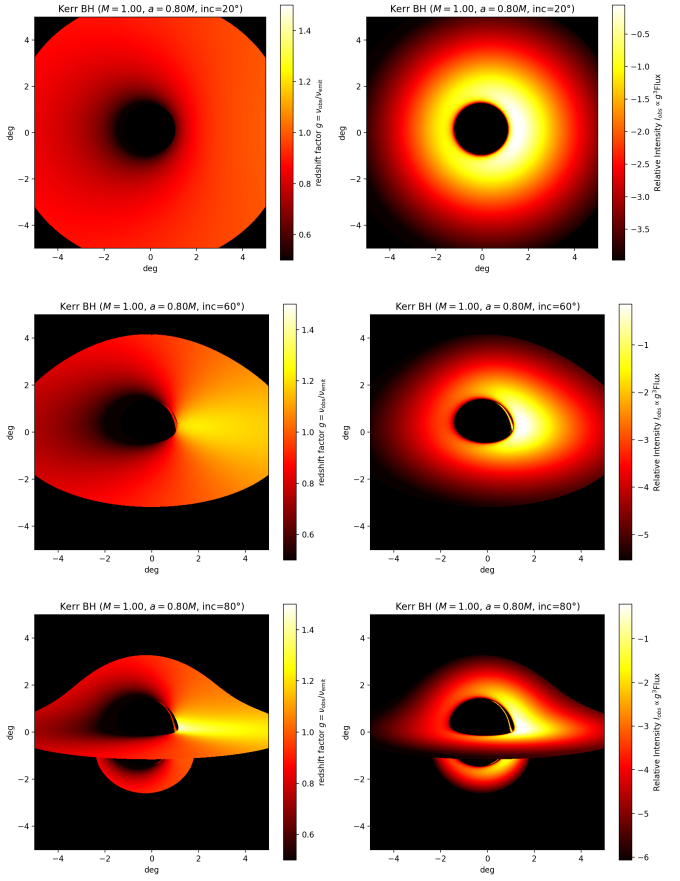


Fig. 4. Inclination dependence of the Kerr black hole image.

by the optically thick disk and appears as a sharp crescent on the inner-right edge of the shadow.

On the retrograde side, the near-critical photon trajectories occur at larger radius. Because their bending is less abrupt in this geometry, these rays travel closer to the equatorial disk plane over a longer path. In the ray-tracing image, they intersect the outer part of the optically thick disk before reaching the observer. As a result, the retrograde-side photon-ring segment is geometrically hidden by the disk.

At low inclination, such as 20°, the observer looks almost along the rotation axis. In this nearly face-on configuration, the direct image of the inner disk is projected onto the same region as the near-critical photon-ring trajectories. Since the disk is optically thick and rays terminate at the first disk intersection, the photon-ring contribution is geometrically screened by the direct disk image. Therefore, the photon ring is not clearly visible at low inclination. This is not because the photon ring disappears, but because the disk screens the near-critical photon trajectories from the observer.

Thus, the inclination changes the photon-ring visibility mainly through geometric projection and disk occlusion. High inclination can expose part of the prograde photon-ring segment, while low inclination causes the ring to be hidden behind the optically thick disk image.

4.3. Spin dependence

Figure 5 shows a spin series with $a = 0.00M$, $0.40M$, and $0.80M$, using the same inclination and disk model. The case $a = 0$ corresponds to the Schwarzschild limit, while larger a represents a more rapidly rotating Kerr black hole.

As the spin increases, frame dragging becomes stronger. This enhances the difference between prograde and retrograde photon trajectories. Increasing spin does not create the photon ring from nothing; instead, it changes which part of the near-critical photon-ring structure remains visible after disk occlusion. As a increases, the visible photon-ring segment becomes more asymmetric and more concentrated on the prograde side.

Spin also changes the position of the ISCO. For a Kerr black hole, the ISCO radius depends on the spin parameter. For prograde orbits, it can be written as

$$r_{\rm ISCO} = M\left[3 + Z_2 \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}\right] \tag{14}$$

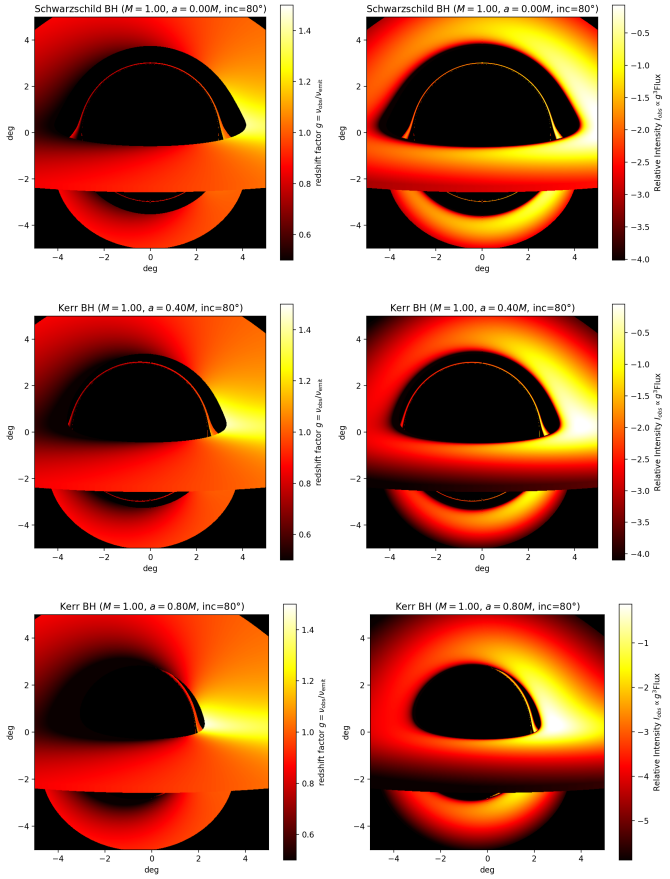


Fig. 5. Spin dependence of the Kerr black hole image.

where

$$Z_1 = 1 + (1 - a_*^2)^{1/3}\left[(1 + a_*)^{1/3} + (1 - a_*)^{1/3}\right] \tag{15}$$

$$Z_2 = \sqrt{3a_*^2 + Z_1^2}, \qquad a_* = \frac{a}{M} \tag{16}$$

This expression shows that, for prograde motion, the ISCO moves inward as the spin increases. Therefore, the inner edge of the accretion disk can approach the black hole more closely. This increases the overlap between the direct disk image and the near-critical photon-ring region.

Since the disk is optically thick, this overlap can hide part of the photon ring. Rays that hit the disk first do not show the higher-order emission behind it. Therefore, in the Kerr case, the photon ring may not appear as a complete circle. Instead, only the segments whose geodesics avoid the disk surface remain visible.

The spin dependence therefore shows that the observed photon-ring segment is not determined only by the existence of near-critical photon trajectories. It is also controlled by whether those trajectories can reach the observer without being intercepted by the optically thick disk. In a rotating Kerr spacetime, frame dragging changes the geometry of these escape paths and produces a one-sided, crescent-like visibility of the photon ring.

4.4. Summary of comparison

The comparisons in Figures 3–5 show that, in these optically thick disk images, the photon-ring segment becomes visible only when the corresponding near-critical photon trajectories are not geometrically screened by the accretion disk. In the Kerr case, frame dragging separates the behavior of prograde and retrograde rays. Prograde rays can bend sharply near the black hole and escape through the inner region, producing the visible crescent-shaped photon-ring segment. Retrograde rays are more likely to be intercepted by the optically thick disk and are therefore hidden in the image.

Inclination also controls the visibility of the ring. At high inclination, the disk geometry allows part of the prograde photon ring to be exposed, while the retrograde side remains hidden. At low inclination, the observer views the system almost face-on, and the direct image of the inner disk geometrically overlaps with the photon-ring region. Therefore, the photon ring is screened almost completely.

Thus, the visibility of the photon ring in these Kerr ray-tracing images should be understood as a geometric effect produced by frame dragging, asymmetric photon geodesics, and occlusion by the optically thick disk. The ring is not absent because the relevant photon trajectories do not exist; rather, it becomes visible or invisible depending on whether those trajectories can escape to the observer without being blocked by the disk.

5. Numerical Validation

To examine the numerical reliability of the ray-tracing calculation, we monitored the null condition of the photon four-momentum along the integrated geodesics. For a photon with four-momentum p^μ, the null condition is

$$g_{\mu\nu}p^\mu p^\nu = 0 \tag{17}$$

In the numerical calculation, however, this contraction is not exactly zero due to finite precision and integration error. Since Eq. (17) is obtained through cancellation among several metric contraction terms, we measured the relative null error

$$\epsilon_{\rm null} = \frac{\left|g_{\mu\nu}p^\mu p^\nu\right|}{\sum_{\mu,\nu}\left|g_{\mu\nu}p^\mu p^\nu\right|} \tag{18}$$

This quantity compares the remaining violation of the null condition with the typical scale of the individual contraction terms.

Figure 6 shows the evolution of $\log_{10}\epsilon_{\rm null}$ for disk-hit rays. The initial null error is typically at the level of 10^{-15}–10^{-18}, indicating that the screen tetrad, the initial photon momentum, and the metric evaluation are mutually consistent to nearly double-precision accuracy. As the rays propagate toward the black hole, the error gradually increases due to accumulated Runge–Kutta integration error. The increase becomes more significant when rays pass through the strong-field region near the inner disk radius.

The right panel of Fig. 6 shows the corresponding radial coordinate evolution, plotted as $\log_{10}(r/r_+)$. The dashed lines indicate the event horizon r_+, the disk inner radius $r_{\rm in}$, and the disk outer radius $r_{\rm out}$. Most disk-hit rays remain outside the horizon and terminate within the disk region. Some rays penetrate inside the disk inner radius before returning to the disk, corresponding to strongly lensed higher-order trajectories associated with the photon-sphere region.

For most disk-hit rays, the relative null error remains around $\epsilon_{\rm null} \sim 10^{-5}$–$10^{-4}$ near the disk crossing. A small number of near-critical rays show larger spikes in the null error, which is expected because such trajectories are highly sensitive to numerical integration errors in the strong-field region. Therefore, the simulation is sufficiently reliable for qualitative image formation and for interpreting the main disk and shadow features. However, the most strongly lensed ring-like trajectories should be interpreted with additional caution or checked using a smaller integration step size.

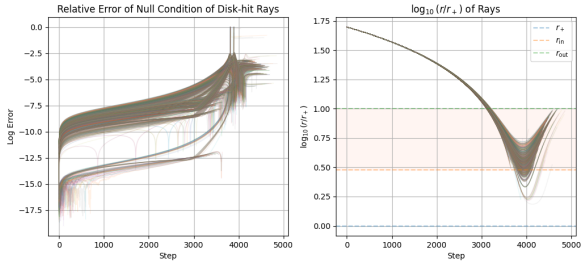


Fig. 6. Numerical validation of the null condition for disk-hit rays.

6. Conclusion

In this project, we developed a ray-tracing simulation of black hole images in Schwarzschild and Kerr spacetimes and used it to interpret several relativistic features of accretion disk images. In the Schwarzschild case, the simulation reproduced the central black hole shadow and photon-ring-like structures associated with strong gravitational lensing near the photon sphere. The comparison between the shadow radius, photon sphere, and ISCO showed that the apparent dark region is determined by

photon capture, while the disk emission structure is controlled by the inner edge of the accretion disk.

We also examined the role of redshift and Doppler effects in shaping the observed intensity distribution. Even in the non-rotating Schwarzschild spacetime, the orbital motion of the disk produces a clear left-right brightness asymmetry through Doppler boosting. This demonstrates that image asymmetry does not necessarily require black hole spin, but can arise naturally from the motion and inclination of the emitting disk.

For Kerr black holes, spin changes the structure of photon trajectories through frame dragging. The comparison of different inclinations and spin parameters showed that the visible photon-ring segment is strongly affected by the geometry of the optically thick disk. In particular, prograde near-critical rays can escape through the inner region and produce a crescent-like photon-ring feature, while retrograde or low-inclination contributions are more easily screened by the disk. Thus, the visibility of the photon ring should be understood not only as a property of the spacetime, but also as a result of disk geometry, inclination, and ray occlusion.

Finally, we checked the numerical reliability of the simulation by monitoring the null condition along disk-hit rays. The initial null error was close to double-precision accuracy, confirming the consistency of the initial tetrad and photon momentum construction. Although the error increases in the strong-field region due to accumulated integration error, the main disk and shadow features remain reliable for qualitative interpretation. More accurate analysis of the most strongly lensed ring-like trajectories would require smaller integration steps or higher-order numerical methods.

Overall, the ray-tracing results demonstrate how black hole shadows, photon-ring-like features, Doppler asymmetry, frame dragging, and disk occlusion emerge from null geodesic motion in curved spacetime. The simulation therefore provides a useful qualitative tool for connecting the mathematical structure of general relativity with observable features of black hole images.

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